Lab 03: Regression Model Comparison using Scikit-Learn

Group 01 – MOSTEFAOUI Manel

November 3, 2025

1 Purpose and Objectives

The purpose of this lab is to explore and compare different regression models in `Scikit-learn`, specifically:

- Multiple Linear Regression
- Polynomial Regression
- Decision Tree Regression

The objective is to predict individual medical insurance costs based on demographic and lifestyle features from the *Insurance (Medical Cost Personal)* dataset from Kaggle. This task is essential to understand how different model complexities capture linear and nonlinear patterns, and how categorical variables such as `smoker` affect predictions.

2 Dataset and Preprocessing

The dataset contains 1338 records with the following features: `age`, `sex`, `bmi`, `children`, `smoker`, `region`, and `charges` (target variable).

Categorical features (`sex`, `smoker`, `region`) were encoded using `OneHotEncoder`. Numerical features were scaled using `StandardScaler`. The dataset was split into 80% training and 20% testing sets.

3 Exploratory Data Analysis

Before modeling, two main relationships were visualized to understand nonlinearity and categorical effects.

3.1 Charges vs BMI (colored by Smoker)

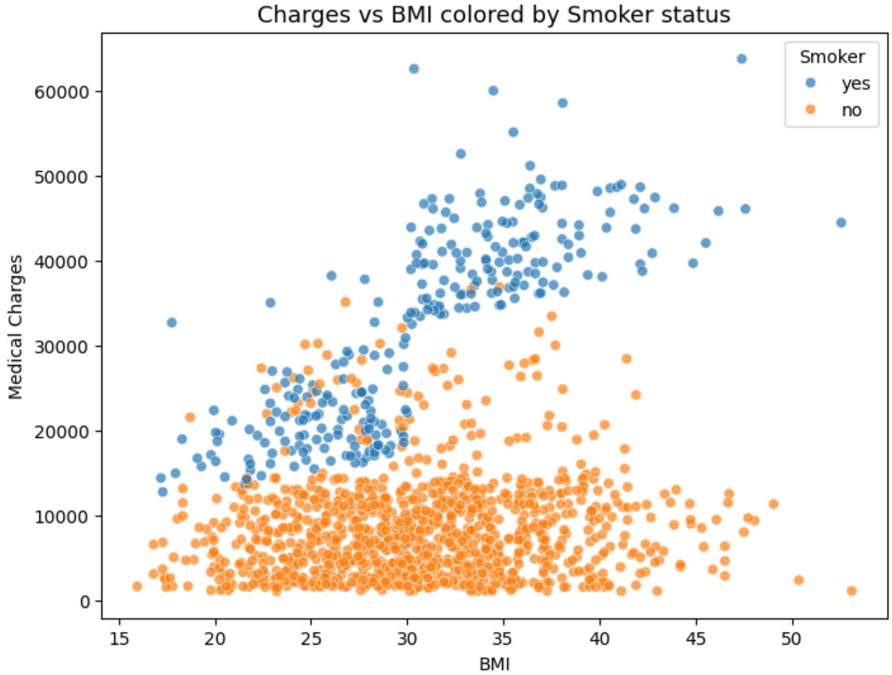

Figure 1: Scatter plot of Charges vs BMI, colored by Smoker status. Smokers show a strong nonlinear rise in charges with BMI.

3.2 Average Charges by Region and Smoker Status

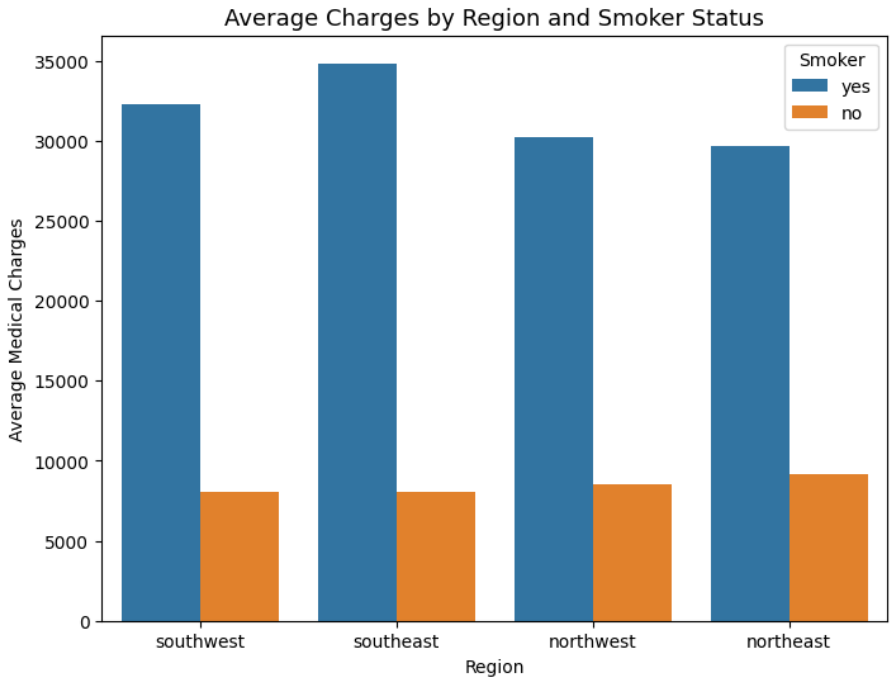

Figure 2: Average charges by region and smoker status. Smokers consistently incur higher charges across all regions.

3.3 Correlation Matrix

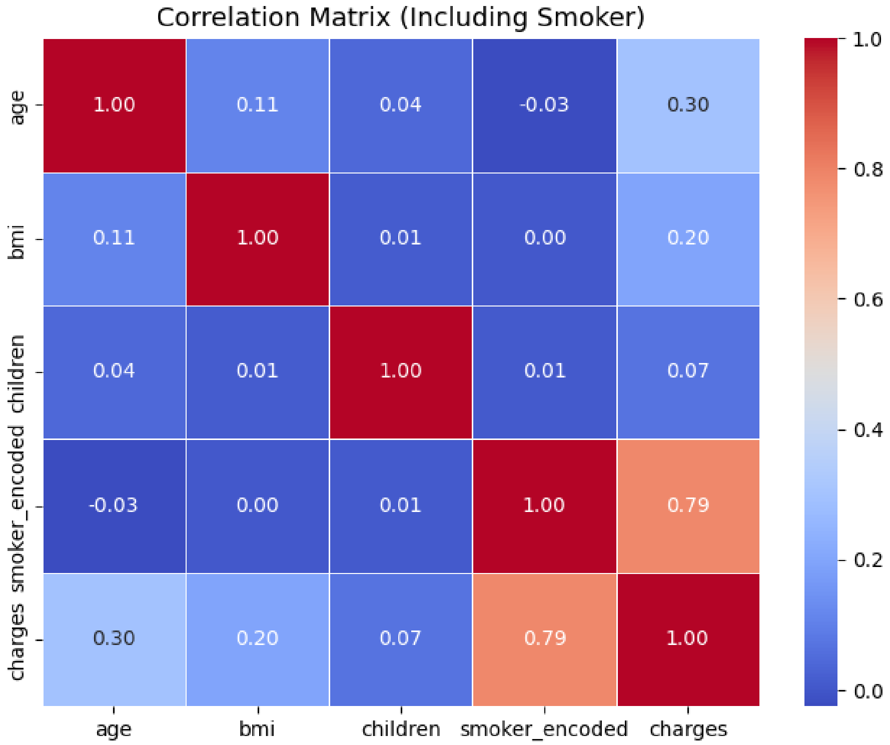

Figure 3: Correlation matrix showing relationships among numeric variables. The strongest correlation is between `smoker` and `charges`.

4 Methodology

Three models were trained using `Scikit-learn Pipelines`:

1. **Multiple Linear Regression:** Fits a linear relationship between predictors and charges.

2. **Polynomial Regression (degree=2):** Adds squared and interaction terms to capture nonlinear effects.

3. **Decision Tree Regression:** Splits the dataset recursively based on feature thresholds to minimize prediction error.

Each model was evaluated using the following metrics:

- Mean Squared Error (MSE)
- Coefficient of Determination (R^2)
- Mean Absolute Error (MAE)

5 Results and Analysis

Model	MSE	R^2	MAE
Linear Regression	3.36e+07	0.7836	4181.19
Polynomial Regression (deg=2)	**2.07e+07**	**0.8666**	**2729.13**
Decision Tree Regression	4.08e+07	0.7372	3114.15

Table 1: Performance comparison of regression models on test data.

Polynomial Regression achieved the best performance with the lowest MSE and highest R^2, capturing nonlinear relationships effectively. Decision Tree Regression showed signs of overfitting—likely due to unlimited depth—while Linear Regression underperformed due to its inability to model nonlinear effects.

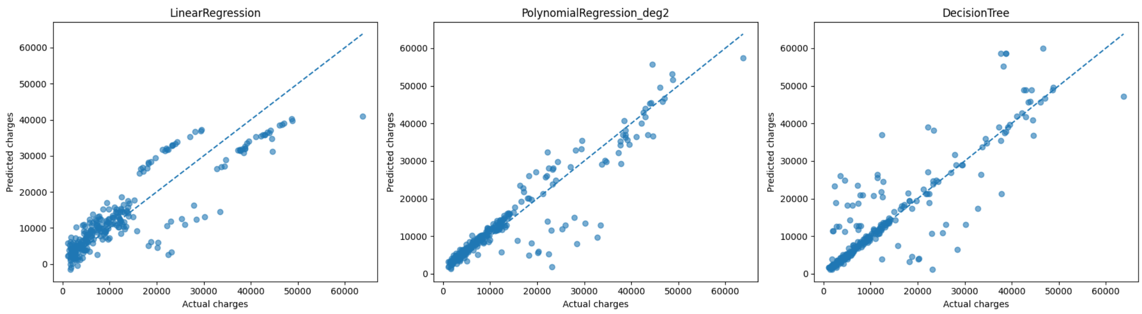

Figure 4: Predicted vs Actual Charges for all models. Polynomial Regression aligns most closely with the ideal diagonal line.

6 Discussion

Nonlinearity: The relationship between BMI, smoking status, and medical charges is strongly nonlinear. Linear models fail to capture the sharp increase in costs for smokers with high BMI, while Polynomial Regression models it more accurately.

Overfitting: The Decision Tree fits the training data perfectly but performs worse on unseen data, showing classic overfitting behavior. Controlling its depth or using ensemble methods (e.g., Random Forest) could improve generalization.

Smoker effect: Being a smoker has the most significant influence on charges. The correlation matrix and feature importances confirm this.

7 Conclusion

Polynomial Regression (degree=2) offers the best balance between accuracy and generalization, achieving $R^2 = 0.8666$ and MAE 2,729. This demonstrates that incorporating nonlinear and interaction terms significantly improves model performance for this dataset. Linear Regression remains a strong baseline for interpretability, while Decision Trees require regularization to avoid overfitting.

Reflection

This lab deepened understanding of model complexity, overfitting, and nonlinear relationships in regression tasks. Through EDA and model comparison, it highlighted how preprocessing, feature encoding, and metric selection directly influence performance.